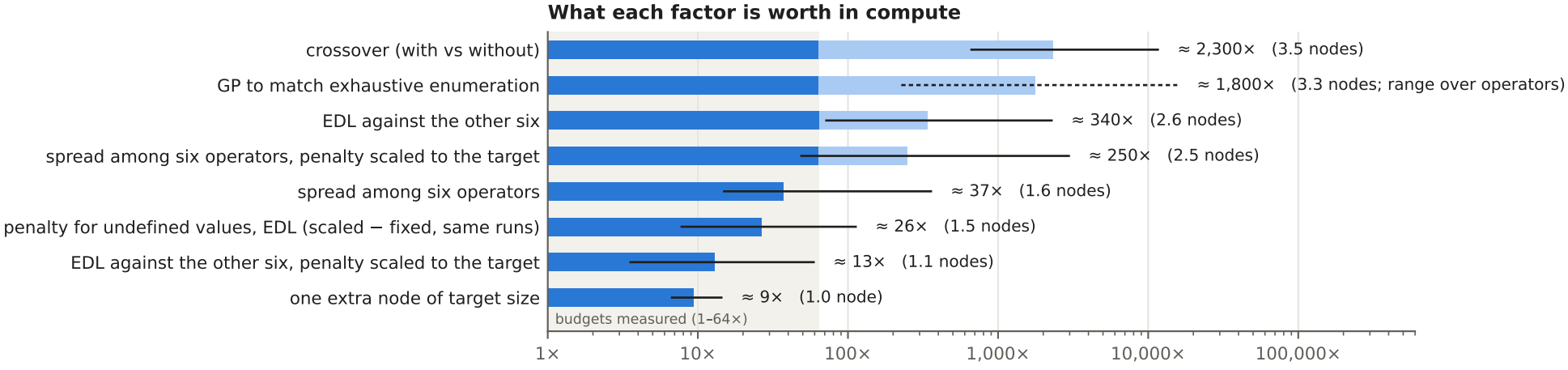
What each factor is worth in compute
crossover (with vs without)
≈ 2,300× (3.5 nodes)
GP to match exhaustive enumeration
≈ 1,800× (3.3 nodes; range over operators)
EDL against the other six
≈ 340× (2.6 nodes)
spread among six operators, penalty scaled to the target
≈ 250× (2.5 nodes)
spread among six operators
≈ 37× (1.6 nodes)
penalty for undefined values, EDL (scaled − fixed, same runs)
≈ 26× (1.5 nodes)
EDL against the other six, penalty scaled to the target
≈ 13× (1.1 nodes)
one extra node of target size
≈ 9× (1.0 node)
budgets measured (1–64×)
1×
10×
100×
1,000×
10,000×
100,000×
equivalent increase in GP's evaluation budget (log scale). Lines: 95% intervals; light bars: extrapolated beyond 64×